

## SATARK - COMPLETE FEATURE OVERVIEW

Problem Statement 26102 MoSPI / DIID 176,925 Real Government Works

### PART 1: THE 7 INDEPENDENT AI SENTINELS

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SENTINEL 1 VIDHI-KAVACH - - Purpose: Legal Rule-Based Fraud Detection

TECH: Indic Lexicon + Rule Matrix

WHAT IT DOES: Checks whether a proposed MPLADS work violates government rules or eligibility conditions.

- Reads the proposed work description.
- Identifies keywords such as temples, trusts, prohibited activities, etc.
- Compares the proposal against predefined MPLADS rules.
- Checks suspicious budget/timing patterns such as March Rush.

DETECTS: Ineligible/prohibited works and statutory rule violations.

BENEFIT: Prevents funds from being sanctioned for legally or procedurally invalid works.

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SENTINEL 2 PUNAR-DRISHTI - Purpose: Duplicate Work Detection

TECH: TF-IDF + Cosine Similarity

WHAT IT DOES: Identifies the same or highly similar work being submitted more than once across schemes.

- Converts work descriptions into numerical representations.
- Compares new proposals with existing proposals across databases.
- Measures textual similarity even when the title has been changed or rephrased.
- Example: Construction of Community Hall vs Development of Village Multipurpose Hall.

DETECTS: Duplicate or suspiciously similar proposals.

BENEFIT: Helps prevent duplicate funding and repeated submission of the same work.

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SENTINEL 3 ARTHA-DARPAN - - Purpose: Cost Price Manipulation Detection

TECH: CPWD DSR Rates + Per-Unit Median Analysis

WHAT IT DOES: Checks whether the estimated project cost appears unusually high compared to government benchmarks.

- Takes project quantities and estimated costs.
- Compares rates against CPWD Delhi Schedule of Rates DSR 2023-24 benchmarks.
- Calculates expected/typical cost ranges for all 37 States/UTs.
- Flags significant upward cost deviations.

DETECTS: Overpricing, inflated estimates, price padding, and abnormally high unit rates.

BENEFIT: Helps identify potential financial irregularities before sanction.

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SENTINEL 4 CHAKRA-VYUH - Purpose: Contractor/Vendor Network Analysis

TECH: Graph Theory + Ego-Graphs Vis.js / Louvain

WHAT IT DOES: Finds suspicious relationships between contractors, vendors, agencies, and projects.

- Represents contractors, vendors, and projects as an interconnected network.
- Creates connections based on common contracts, locations, and officials.
- Looks for highly connected or repeatedly appearing entities using Herfindahl-Hirschman Index HHI.

DETECTS: Contractor concentration, vendor monopolies, cartel-like networks, and repeated contractor-project relationships.

BENEFIT: Reveals hidden syndicate patterns that are difficult to identify by checking individual records.

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SENTINEL 5 BHU-DRISHTI - Purpose: Geospatial Physical Verification

TECH: GIS + Remote Sensing + Geotagging Leaflet / Esri Satellite

WHAT IT DOES: Checks whether a sanctioned asset/work actually exists at the claimed geographic location.

- Uses project coordinates and GPS geotags.
- Maps the project location using GIS satellite layers.
- Compares available geographic and satellite evidence.
- Identifies suspicious or unverifiable locations water bodies, forests, barren land.

DETECTS: Ghost assets, incorrect coordinates, location mismatches, and non-existent works.

BENEFIT: Adds a physical verification layer to digital records.

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SENTINEL 6 SANKHYA-SATYA - Purpose: Tender Financial Pattern Detection

TECH: Benfords Law + Smurfing Rules

WHAT IT DOES: Looks for suspicious tender patterns that indicate attempts to avoid open public scrutiny.

- Analyzes financial amounts and transaction patterns.
- Detects unusual numerical distributions on leading digits 1-9.
- Looks for multiple smaller transactions/works that collectively represent a larger suspicious amount.

DETECTS: Tender splitting e.g. 4.95L bills, artificially small project values, suspicious financial distributions, and threshold-avoidance.

BENEFIT: Helps identify financial patterns that may not be obvious from individual transactions.

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SENTINEL 7 VIBHED-NETRA - Purpose: Advanced Multi-Parameter Anomaly Detection

TECH: 12D Isolation Forest Machine Learning

WHAT IT DOES: Detects complex anomalies by considering multiple project parameters simultaneously.

- Takes 12 dimensions: cost, date, location, vendor, project type, execution time, and administrative parameters.
- Machine learning identifies non-linear combinations that look unusual compared with normal projects.

DETECTS: Complex multi-dimensional anomalies that individual rule-based checks may miss.

BENEFIT: Provides an additional AI/ML-based fraud detection layer.

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## PART 2: CORE SATARK ARCHITECTURE 6 MODULES

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Module 8 SATARK-DRISHTI - - Purpose: Unified Risk Scoring

TECH: Composite Risk Scorer 0100

WHAT IT DOES: Combines the results of all 7 Sentinel engines into a single dashboard risk score.

- Collects alerts from all 7 Sentinels.
- Assigns weights and severity to different risks.
- Produces a single overall composite risk score 0-30 Low, 31-60 Moderate, 61-80 High, 81-100 Critical.
- Estimates potential project delay and risk trends.

BENEFIT: Gives officers a single dashboard-level risk assessment instead of checking seven systems separately.

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Module 9 SATARK-SAMVAAD - - Purpose: AI-Powered Government Data Assistant

TECH: GenAI Natural Language Copilot

WHAT IT DOES: Allows officers to ask questions about government and project records in plain language.

- Officer asks a normal-language question in English or Hindi.
- AI interprets the intent, location, budget, and sentinel filter.
- Searches 176,925 MoSPI government records in real time.
- Returns a concise answer with supporting evidence table.

BENEFIT: Makes large government datasets easier for non-technical users to analyze.

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Module 10 PRASHNA-KAVACH - - Purpose: Explainable AI / Reason Behind the Alert

TECH: SHAP Explainable AI Drawer

WHAT IT DOES: Explains why SATARK flagged a project with transparent evidence attribution.

- Breaks down factors contributing to a risk score.
- Shows which variables had the strongest influence.
- Example: High cost deviation +25, Suspicious vendor network +20, Location anomaly +15.
- Answers: Why was this project flagged

BENEFIT: Makes AI decisions transparent and easier for auditors and officers to trust and verify.

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Module 11 SATARK-SIMULATION - - Purpose: Pre-Sanction What-If Testing

TECH: Real-Time Sub-15ms Sandbox Environment

WHAT IT DOES: Allows officials to test changes to a proposal in 15ms before approving it.

- Creates a simulated copy of the proposal.

- Officer changes parameters such as cost, vendor, or project details.
- SATARK immediately recalculates the risk across all 7 sentinels.
- Example: 50 Lakh estimate - 40 Lakh estimate - risk score recalculates in 2ms.

BENEFIT: Supports better decisions before funds are sanctioned.

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Module 12 BHAVISHYA-REKHA - - Purpose: Future Risk Trend Forecasting

TECH: Linear Regression Time-Series Forecasting

WHAT IT DOES: Predicts potential future risk trends and March Rush surges from historical project data.

- Analyzes historical project patterns.
- Identifies trends such as increasing March-Rush activity.
- Forecasts future quarterly risk patterns.
- Highlights seasonal risk spikes and high-risk periods.

BENEFIT: Moves SATARK from reactive detection to proactive prevention.

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Module 13 SATARK-KARYA - - Purpose: Automated Vigilance Documentation

TECH: GFR-19A Memo Builder + SHA-256 Hashing

WHAT IT DOES: Converts detected violations into formal legal documentation for further administrative action.

- Takes investigation and alert details.
- Generates a structured Form GFR-19A vigilance memo/report.
- Includes supporting evidence and 5-point engineer inspection checklist.
- Uses SHA-256 hashing for document integrity.

BENEFIT: Reduces manual paperwork and creates a consistent audit and action trail.

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